



**SIMATS**  
ENGINEERING



**SIMATS**  
Saveetha Institute of Medical And Technical Sciences  
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

## **Undergraduate Internship Report**

Program: B-Tech

Department: Computer Science Engineering

Academic Year: 2023 - 2027

Internship Duration: 1st Aug 2025 to 31st Oct 2025

Name of Student: Chunduri Naga Sai Srikanth

Register Number: 192311159

Batch: 2023–2027

Internship Organization: ONEYES INFOTECH

SOLUTIONS Supervisor in Company: A. SARANYA

[HR Director]

## 1. Internship Completion Certificate



# CERTIFICATE OF EXCELLENCE

This Certificate is proudly awarded to

**CHUNDURI NAGA SAI SRIKANTH**

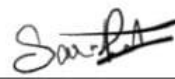
**B.E-CSE**

**SAVEETHA SCHOOL OF ENGINEERING**

For Completing **Python Developer** Internship Program at **OneYes Infotech Solutions Pvt. Ltd.** during the period of **1st August 2025 to 31st October 2025.**

  
**A.SARANYA**  
HR DIRECTOR



  
**M.SARATH KUMAR**  
HR MANAGER

## **2.Student Declaration**

I, CH. Naga Sai Srikanth , a student of Bachelor of Engineering (B.E) in the Department of CSE solemnly affirm that the internship experience documented in this report reflects my direct involvement in a formal organizational setting, actively engaged in industry-relevant operations.

During the internship period from 01/08/25 to 31/10/25. I was entrusted with tasks and responsibilities that were integral to the company's ongoing functions. The insights and observations included herein are a true representation of the work culture, technical exposure, and professional interactions I encountered.

This report has been prepared based on actual experiences and contributions during my time with the organization. I have abided by the professional conduct expected of an intern and ensured that the contents of this document are factual, original, and aligned with the intended academic and industrial learning objectives.

CH. Naga Sai Srikanth

Date: 31/10/25

## **2. Company Overview**

- Name of the Company: Oneyes infotech solutions
- Nature of Business/Industry Sector: Information Technology
- Brief History & Overview: newer private limited company (registered June 2024 in Chennai) and an established firm (founded around 2012)
- Location(s): Chennai, Tamil Nadu
- Products/Services Offered: python developer.

### 3. Internship Role and Objectives

- **Internship Title/Designation:** AI Intern
- **Department/Division Assigned:** R&D (Research & Development) / Software Development Team
- **Reporting Manager/Supervisor:** Sarath Murugadoss

#### **Internship Objectives (as provided by the company):**

1. To design and develop an AI-powered smart resume analyzer that automatically parses resumes and extracts key information using Natural Language Processing (NLP).
  2. To build a job matching system that compares resumes with job descriptions and provides accurate match scores using machine learning techniques.
  3. To develop a scalable platform that assists both job seekers and recruiters with resume analysis, candidate shortlisting, and improvement suggestions.
- 

### 4. Work Description / Tasks Assigned

- Designed the overall system architecture for the AI-powered resume analyzer platform.
  - Developed modules for resume upload, parsing, and structured data extraction.
  - Implemented NLP pipelines for entity recognition (skills, education, experience).
  - Built a Resume Matching Engine using:
    - TF-IDF and cosine similarity
    - Sentence-BERT embeddings for semantic matching
  - Developed dashboards for job seekers and recruiters.
  - Implemented AI-based feedback system for resume improvement suggestions.
  - Generated PDF reports for match scores and analysis.
  - Documented system performance, accuracy, and workflow.
- 

#### **Tools / Technologies / Software Used**

- **Programming Languages:** Python, JavaScript
  - **Frontend:** React.js, Material UI
  - **Backend:** Django REST Framework (DRF)
  - **NLP & ML:** spaCy, Scikit-learn, Sentence-BERT
  - **Resume Parsing:** PyMuPDF, python-docx, regex
  - **Database:** PostgreSQL
  - **Storage:** AWS S3
  - **Deployment:** Docker, AWS EC2, NGINX
  - **Other Tools:** Redis, Celery, GitHub Actions
-

## **Real-Time Project Participated In**

### **Project Title: AI-Powered Smart Resume Analyzer for Job Matching**

#### **Details:**

This project focuses on building an intelligent system that analyzes resumes and matches them with job descriptions using AI and NLP techniques. It helps job seekers improve their resumes and assists recruiters in selecting the best candidates efficiently.

---

#### **Modules:**

##### **1. Resume Upload & Parsing**

- Users upload resumes in PDF, DOCX, or TXT formats.
- Extracts details like name, skills, education, and experience.
- Uses NLP (spaCy NER) for structured data extraction.

##### **2. Job Description Processing**

- Recruiters post job descriptions.
- Extracts required skills and qualifications.
- Uses NLP preprocessing and embedding techniques.

##### **3. Resume Matching Engine**

- Compares resumes with job descriptions.
- Uses TF-IDF and BERT embeddings.
- Generates match score (0–100%) and keyword analysis.

##### **4. Feedback & Recommendation System**

- Suggests missing skills and improvements.
- Recommends courses for skill enhancement.
- Provides resume quality insights.

##### **5. Dashboard & Reports**

- Displays match scores and analytics.
- Generates downloadable PDF reports.
- Visualizes results using charts and graphs.

##### **6. Admin Control Panel**

- Monitors users and system activity.
  - Provides analytics and moderation tools.
- 

#### **Meetings / Interactions Attended**

- Weekly progress review meetings with the technical lead.
- Design discussions on resume parsing and matching algorithms.
- Technical sessions on improving NLP accuracy and performance.

- Code review meetings for modular and scalable implementation.
- 

### **Key Responsibilities & Daily Activities**

- Developing backend APIs for resume parsing and job matching.
  - Implementing NLP models for entity extraction and similarity matching.
  - Designing dashboards for users and recruiters.
  - Testing system performance and accuracy across datasets.
  - Debugging and optimizing matching algorithms.
- 

## **5. Learning Outcomes**

### **Technical:**

- Gained practical knowledge of NLP techniques like tokenization, NER, and embeddings.
- Learned resume parsing and job matching using AI models.
- Hands-on experience with real-time deployment using AWS and Docker.

### **Soft Skills:**

- Problem-solving and analytical thinking.
- Technical documentation and teamwork.
- Time management and task prioritization.

### **Understanding of Industrial Workflows:**

- Learned importance of preprocessing (text cleaning, normalization).
- Understood modular architecture and scalable system design.
- Gained experience in CI/CD and cloud deployment.

### **Contribution to Team/Project:**

- Successfully developed a functional Resume Analyzer system.
- Implemented accurate job matching and feedback mechanisms.
- Improved system efficiency and usability.

### **Comparison with Academic Learning:**

- Academic learning focused on theory (AI, ML, NLP).
  - This project provided real-world implementation experience and deployment knowledge.
- 

## **6. Challenges Faced and Solutions**

### **Technical Challenges:**

- **Resume Parsing Complexity:** Different resume formats caused extraction errors.
- **NLP Limitations:** Difficulty in accurately identifying skills and entities.
- **Matching Accuracy:** Ensuring precise similarity between resumes and job descriptions.
- **System Performance:** Handling large datasets efficiently.

- **Security & Privacy:** Protecting user data and uploaded resumes.
- 

#### **How They Were Resolved:**

- **Improved Parsing Techniques:** Used regex + NLP for better extraction accuracy.
  - **Enhanced NLP Pipeline:** Applied NER models and embedding techniques.
  - **Optimized Matching Algorithms:** Combined TF-IDF and BERT for better results.
  - **Efficient System Design:** Used Celery and Redis for asynchronous processing.
  - **Secure Implementation:** Applied authentication, encryption, and secure storage.
- 

#### **Feedback from Supervisor:**

“Appreciated for successfully developing an AI-driven resume analyzer with efficient job matching capabilities. Demonstrated strong knowledge in NLP, machine learning, and system design with a scalable and user-friendly approach.”

---

### **7. Reflection and Conclusion**

#### **Overall Experience:**

The internship provided hands-on experience in developing an AI-based resume analysis and job matching system. It bridged the gap between theoretical AI concepts and real-world applications.

#### **Career Relevance:**

- Strengthened interest in AI, NLP, and backend development.
  - Useful for careers in AI Engineering, Data Science, and Software Development.
- 

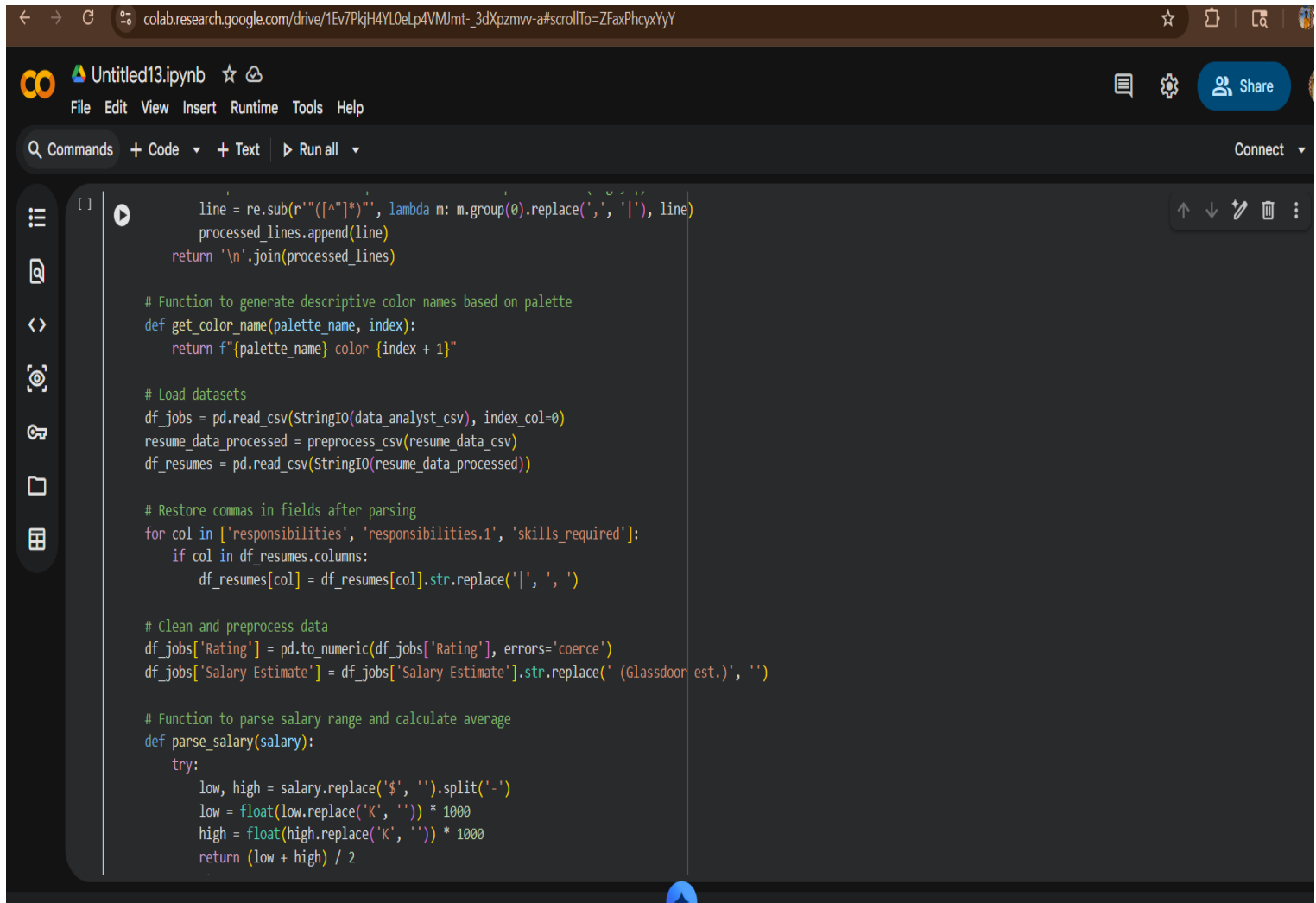
#### **Suggestions for Future Interns:**

- Focus on core concepts like Python, NLP, and Machine Learning.
  - Practice real-time project development and API integration.
  - Work on optimizing algorithms and handling edge cases.
- 

#### **Suggestions for College/Department:**

- Introduce more practical AI and NLP-based projects.
- Encourage cloud deployment and real-world system design.
- Promote interdisciplinary learning (AI + Web + Cloud).

## 8. Appendices



The screenshot shows a Google Colab notebook interface. The browser address bar at the top displays the URL: `colab.research.google.com/drive/1Ev7PkjH4YL0eLp4VMJmt_3dXpzmwv-a#scrollTo=ZFaxPhcyxYyY`. The notebook is titled "Untitled13.ipynb" and has a menu bar with options: File, Edit, View, Insert, Runtime, Tools, and Help. Below the menu bar is a toolbar with icons for Commands, Code, Text, Run all, and Connect. The main area of the notebook contains a single code cell with the following Python code:

```
[ ] ▶ line = re.sub(r'("[^"]*)"|', lambda m: m.group(0).replace(',', '|'), line)
      processed_lines.append(line)
      return '\n'.join(processed_lines)

# Function to generate descriptive color names based on palette
def get_color_name(palette_name, index):
    return f"{palette_name} color {index + 1}"

# Load datasets
df_jobs = pd.read_csv(StringIO(data_analyst_csv), index_col=0)
resume_data_processed = preprocess_csv(resume_data_csv)
df_resumes = pd.read_csv(StringIO(resume_data_processed))

# Restore commas in fields after parsing
for col in ['responsibilities', 'responsibilities.1', 'skills_required']:
    if col in df_resumes.columns:
        df_resumes[col] = df_resumes[col].str.replace('|', ', ')

# Clean and preprocess data
df_jobs['Rating'] = pd.to_numeric(df_jobs['Rating'], errors='coerce')
df_jobs['Salary Estimate'] = df_jobs['Salary Estimate'].str.replace(' (Glassdoor est.)', '')

# Function to parse salary range and calculate average
def parse_salary(salary):
    try:
        low, high = salary.replace('$', '').split('-')
        low = float(low.replace('K', '')) * 1000
        high = float(high.replace('K', '')) * 1000
        return (low + high) / 2
```

1:59

VoLTE 52.35 KB/s 5G



GPS Map Camera



### **3. Faculty Mentor's Remarks**

This is to certify that the internship report submitted by CH. Naga Sai Srikanth has been reviewed and it is evident that the internship was undertaken at a bona fide industry or company. The work documented is satisfactory and relevant to the student's field of study.

Faculty Name & Designation

Signature:

Date: